

Garv Arora

Computer Science Engineer | Intelligent Robotic Systems & Software

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Education

Vellore Institute of Technology (VIT)

Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) | CGPA: 8.36 / 10.0

Vellore, India

Aug 2023 – Jul 2027

Gems International School (CBSE)

Class XII: 76.4% | Class X: 89.3%

Gurugram, India

Nov 2016 – Apr 2023

Work Experience

LG Soft India (LGSi) – HS/ES India Lab

Bengaluru, India

Living Solution Control Development Intern

Jun 2026

- Engineered C/C++ embedded control logic and state machine handlers for living solution control modules, verifying execution timing on target hardware; work certified as “Technically Competent” by LGSi HR/Engineering management.

Samsung PRISM

Remote

Data Intelligence Agent Developer (Internship)

Aug 2025 – Present

- Engineered multi-agent LangGraph data engine (FastAPI, Redis, PostgreSQL, Qdrant) automating discovery and 20-stage analysis of dataset corpora; integrated spaCy NLP, Presidio PII anonymization, and MemorySaver HITL pause.
- Built automated dataset validation microservices enforcing SPDX license compliance and BERTopic semantic clustering across multi-GB data repositories.

Wissen Baum Engineering Solutions LLP

Pune, India

Software Automation Intern

May 2025 – Jul 2025

- Built Python BDD test automation framework (Gherkin/Behave, pytest, Playwright), parallelizing GitLab CI jobs to cut pipeline execution time from ~10m to ~6m (40% speedup) and reducing manual validation effort by 80%+.

- Developed pandas/NumPy data processing scripts processing 10,000+ test case records per run, automating test report generation.

SEDS India (SEDS VIT)

Vellore, India

Research & Development Lead (Promoted from Developer)

Apr 2024 – Present

- Promoted to R&D Lead directing 2 space robotics teams (50+ engineers) across Team Vyadh (IRC) and Team Ardra (ISDC); conducted architecture reviews and debugging, resolving 10+ critical pre-competition issues.
- Architected ROS2 Nav2 & RTAB-Map 3D Visual SLAM on Intel RealSense D455, generating real-time elevation maps and cutting manual teleoperation time by 70% in competition traversals.
- Slashed CV latency by ~60% and boosted throughput 4x using CUDA GPU kernels, YOLOv8, ArUco/AprilTag tracking, and V4L2 zero-copy buffer tuning across 4 camera streams.
- Flashed Micro-ROS firmware onto ESP32 microcontrollers, fusing wheel encoders and IMU with an Extended Kalman Filter (EKF) to cut odometry drift by ~40% (sub-10 cm accuracy).

Projects

HayaiOS – Preemptive Bare-Metal RTOS Kernel | C, ARM Assembly, ARM Cortex-M4

2026

- Implemented preemptive scheduler (1 ms tick) with context switching in <50 ARM assembly instructions, mutex/semaphore IPC primitives, and a hardware abstraction layer (HAL) isolating register-level I/O on ARM Cortex-M4.

- Benchmarked scheduler performance across 8 concurrent tasks, achieving zero priority inversion and <2 μ s context switch latency.

Captivity CLI | Rust, Python, Linux D-Bus, systemd, Socket IPC | pypi.org/project/captivity-cli

2026

- Built autonomous captive portal WiFi daemon featuring <50ms HTTP probing, systemd integration, D-Bus NetworkManager monitoring, keyring encryption, and dual Python-Rust TCP socket IPC.

teleop-cursor | ROS 2, Python, rclpy, geometry_msgs/Twist | pypi.org/project/teleop-cursor

2026

- Developed zero-hardware mouse cursor teleoperation node for ROS 2, converting screen displacement vectors into smooth velocity commands (*geometry_msgs/Twist*) at 10 Hz callback frequency for TurtleBot3/Gazebo.

3D Reconstruction – IMU-Enhanced KinectFusion | C++, OpenCV, Intel RealSense SDK

2025

- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and 6-DOF IMU data via complementary filter, reducing tracking failures by ~50% across fast-motion trajectories.

Patents

Patent Published (App IN202641072249 A1): Autonomous Radar-Guided Survivor Detection and Navigation System — Fused

24GHz FMCW mmWave radar, ultrasonic array, and IMU dead-reckoning on dual-core ESP32; formulated confidence grid decay

$C = \max(0, C - \beta \Delta t)$ for survivor breathing detection under rubble.

Achievements & Awards

Amazon ML Summer School 2026: Selected among Top 3,000 out of 1,30,000+ applicants across India (~2.3% acceptance rate).

International Rover Challenge (IRC): Ranked **13th Globally (2025)** and **17th Globally (2026)** with Team Vyadh (SEDS VIT).

Certifications

AWS Certifications: AWS Certified AI Practitioner | AWS Certified Cloud Practitioner

Oracle Cloud Certifications: Certified Data Science Professional | Certified Generative AI Professional

Technical Skills

Languages: C, C++, Python, Rust, ARM Assembly, Java, JavaScript, SQL, Bash, Verilog

Robotics & Autonomous Systems: ROS2 (Nav2, MoveIt2), RTAB-Map (3D SLAM), Micro-ROS, EKF/UKF, PID/MPC, Sensor Fusion, mmWave Radar

Perception & Machine Learning: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, Transformers

Embedded & Systems: ARM Cortex-M, ESP32, FreeRTOS, I2C/SPI/UART, Bare-Metal C/Assembly, Linux Kernel/D-Bus, Docker, Git, CI/CD, AWS